ZHANG MENGKE

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EDUCATION

Beijing University of Posts and Telecommunications (BUPT) *Sep 2023 - Present*
Bachelor of Internet of Things Engineering
3.62/4.0 (Average Score: 89.5/100)
Honors: Honorable Mention, Mathematical Contest in Modeling (MCM/ICM); Third Prize, 2025 ICCIP Information Innovation Competition; Third Prize, Asia-Pacific Mathematical Modeling Competition

ACADEMIC EXPERIENCE

OpenDepth: Robust Open-World Monocular Depth Estimation ACM MM 2026 (rating 4/5)
First Author — Advisor: Prof. Anlong Ming(BUPT) *Generative Models, Depth Estimation*

- **I2I Editing Priors**: Leveraged I2I editing priors to resolve structural hallucinations in text-to-image models, proposing the OpenDepth framework for pixel-level integrity under extreme environments.
- **Intra-Domain Awareness**: Designed an Intra-Domain Awareness mechanism in shallow DiT blocks with a unified RGB latent reconstruction task. Achieved zero-shot SOTA on extreme weather benchmarks using only 79K samples.

Prior-Guided Single-Stream Architecture for Multimodal Remote Sensing ACM MM 2026 (rating 3.4/5)
First Author — Advisor: Prof. Junli Yang(BUPT) *Remote Sensing, Linear Attention*

- **Single-Stream Architecture**: Proposed a prior-guided architecture, decoupling DSM into lightweight geometric priors to reduce cross-modal redundancy while preserving fine-grained semantics.
- **GS-MoE Module**: Designed a GS-MoE module (8-select-4 sparse routing) to prune directional noise, coupled with an FCG module. Achieved 68.51% mIoU on US3D and 85.13% on Vaihingen.

Resolving "Semantic Dominance" via Acoustic-Semantic Decoupling Interspeech 2026 (Accepted)
First Author — Advisor: Prof. Chao Liu(QMUL) *Audio LLMs, Affective Computing*

- **Adversarial Benchmark**: Created the **ASPIRE** benchmark for "tone-text conflict" and proposed SOP/LDD metrics to quantitatively expose semantic hegemony in multimodal LLMs.
- **Decoupling Architecture**: Developed **ACR-Net** via LoRA and contrastive loss to project features into orthogonal latent spaces, outperforming SOTAs and mitigating inaccurate emotional realization.

A Unified Image Deraining System for Real-World Scenarios ICASSP 2026 (Accepted)
Second Author — Advisor: Prof. Weidong Gao(BUPT) *Image Restoration, Multi-Expert Networks*

- **Dataset Generation**: Built **Hybrid-Rain10K** via Blender (10,000 pairs) with precise control over rain density, angle, and transparency across three degradation types.
- **UD2MoE Formulation**: Proposed UD2MoE, an asymmetric U-shape model with a Dual-MoE strategy (Encoder Soft-MoE + Decoder Hard-MoE) to adaptively process mixed rain degradation patterns.

INTERNSHIP EXPERIENCE

Xiaomi Group *Mar 2026 - Present*
Autonomous Driving Algorithm Intern — Mentor: Dr. Naiyan Wang *Beijing, China*

Occupancy (OCC) Perception & Penalty: Developed a fast NPC-centric ray-casting module. Injected OCC features via 1D-CNN/cross-attention and designed an OBB SDF-based collision loss to drastically reduce static obstacle collisions.

Multimodal Trajectory Generation (Mimir): Spearheaded a Flow Matching paradigm for multi-intent ego-trajectories, optimized via Top-k WTA loss and Frenet (s, l) decoupled supervision. Deployed using Euler ODE integration and K-means++ for diverse, real-time inference.

PROFESSIONAL SKILLS

- **Programming Languages:** Python, C/C++, Java, MATLAB.
- **Deep Learning Frameworks & Architectures:** PyTorch, TensorFlow; Familiar with Transformer, State Space Models , and Diffusion Models.
- **CV & Data Processing Libraries:** OpenCV, PIL/Pillow, scikit-image, NumPy, Pandas, SciPy, scikit-learn.